

This file is the canonical statement of **why** conception-space exists and **what it is trying to do**. It is not a roadmap, not a feature list, and not a how-to. It is the **intent** — the underlying purpose that any specific feature, design decision, or refactor should be in service of.

Use it like this:

- When proposing a new feature, check whether it serves the intent below. If it doesn't, it probably doesn't belong.
- When choosing between two implementations, prefer the one that better preserves the intent.
- When working becomes confused about “what is this even for,” return here.
- When the work surfaces a new design principle that follows from the intent, add it as an axiom below so future work inherits it.

THE INTENTS

conception-space serves four complementary intents. Different facets of the same purpose, mutually reinforcing — the tensions between them are themselves design-revealing.

Parallel perception of relationships

The perceptual-channel argument.

1

Hand-authoring as sense-making

Placement as the author's reasoning.

2

Sharing a mental model

The file as a transmissible argument.

3

A persistent thinking artifact

A long-lived companion to ongoing thought.

4

1. Parallel perception of relationships

conception-space is for parallel perception of relationships.

LINEAR

read one at a time

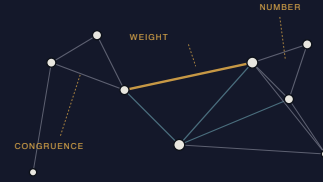
A → B

A → C

PARALLEL

seen at a glance

B → D
 C → D
 D → E
 E → F
 F → A
 G → C
 H → E
 ↓ you read down
 ...



A list, an index, or an appendix forces relationships through a linear perceptual channel — you read item by item, holding the previous in working memory while you read the next, building the structure incrementally and forgetting most of it by the end. Even with a great index, “what is the shape of these 200 relationships?” is unanswerable from the index itself; you have to construct that shape in your head, one relationship at a time.

A `.cns` space pushes the same information through a parallel perceptual channel — your visual system, which is built exactly for this. The moment you look at the scene, three things become available **at once**, without sequential effort:



Number

How many relationships exist — read by edge density and node count at a glance.

“This region is busy; that one is sparse.”



Weight

The strength or importance of relationships — read by edge thickness, colour, label, shape.

“These connections are heavy; those are incidental.”



Congruence

How relationships align, cluster, and point — read by spatial proximity and shared direction.

“These two clusters point at each other; this one is an outlier.”

The user does not recall the structure — they see it. Reasoning is not sequential reconstruction; it is pattern recognition over a structure presented as a whole. That is a fundamentally higher-bandwidth way to grasp relational data than any linear medium can be, and it is the reason this tool exists.

Spatial memorability (Method-of-Loci-style “I remember Mars because it’s over there”) is a secondary effect that follows from the placement being stable across sessions. Useful, but not the primary thing.

2. Hand-authoring as sense-making

The act of placing a node is an act of reasoning. The author decides where it goes — and that decision is informed by their understanding of how it relates to the others. As more nodes are placed, the understanding sharpens; as the understanding sharpens, nodes move. The file is the residue of that thinking.

This is why hand-placement is non-negotiable (Axiom 1): if a layout algorithm overwrites the positions, it overwrites the reasoning. The author is not just describing a graph; they are thinking about it in 3D space. A blank `.cns` file is an invitation to think. A finished one is the shape of a thought.

3. Sharing a mental model

Once a `.cns` file exists, anyone with the viewer can see the author's argument directly. The spatial layout is the author's understanding made transmissible. Reading someone else's `.cns` is not reading their summary of the concepts; it is perceiving the structure they perceived.

Two consequences. The file is legible across people with no narrator — the shape stands by itself. And an author can use the file to teach — laying out the topology of an idea and walking a viewer through it, knowing the viewer is taking in the structure simultaneously, not sequentially. This is the social complement to perception: parallel perception of someone else's relational argument.

4. A persistent thinking artifact

A `.cns` file lives across sessions. The author can leave it and come back. They can evolve it as their understanding evolves. They can revisit a layout from last year and ask "what was I thinking when I put that there?" — and often the placement itself encodes the answer.

The viewer's bookmarks (Axiom 4) operate on the short time-scale — saved viewpoints into a file for a particular act of perception. The file itself is the longer time-scale: a thinking-vehicle that outlasts any single sitting. A `.cns` is not a chart that captures a moment; it is a workshop the author keeps returning to.

How the four reinforce each other

Each intent strengthens the others. Sense-making depends on perception: the author can only reason about the layout if they can see it parallel-channel as they build it. Perception depends on sense-making: the layout has its high-bandwidth structure only because the author placed it deliberately. Sharing depends on both: the file is transmissible only because the author's argument is encoded in the layout and readable in parallel by the viewer. Persistence is the time-axis that lets all three accumulate value: a one-off arrangement is a sketch; a returned-to arrangement is a body of work.

Tensions between them are design-revealing. For example: a feature that improves perception by auto-rounding edge paths for visual cleanliness would damage sense-making if it overrides what the author drew. Both intents are real; the tension forces the design to be specific about what it favours and why.

AXIOMS

Six design constraints that follow from the intents above. Load-bearing in past decisions; load-bearing for future ones.

1 Placement is argument.

When the author places node A near node B and far from C, that placement **is** the claim “A and B are related; C is not.” The space is the author’s understanding of the relationships made directly perceptible. Anything that overwrites placement (auto-layout, repositioning on resize, etc.) erases the author’s argument. Hand-placement is non-negotiable.

2 Hierarchical grouping carries meta-relationships.

A `node_group` says “these N things are one thing at the next scale up.” The group’s own position then encodes how that aggregate relates to other aggregates. A `.cns` file is a multi-scale relational argument: each level reveals a different layer of structure. Treat group-level visuals (shell, tube frame, label) as carrying real reasoning content, not decoration.

3 Smooth motion is load-bearing, not polish.

A snap, spin, or teleport breaks the perceptual channel mid-act. The user stops perceiving relationships and starts re-orienting from scratch. Smooth flight lets the user **sweep** across many relationships in a single continuous perceptual act, picking up patterns the static view can’t show (“the edges from this cluster all bend toward that region”). Camera transitions are part of the medium, not decoration on top of it.

4 Bookmarks are saved arguments.

A bookmark is a curated viewpoint that frames a specific relationship-pattern for a specific perceptual act. The viewpoint **is** the argument. Bookmarks deserve first-class treatment (naming, organisation, easy travel) because each one is a piece of authored sense-making.

5 Edges carry as much reasoning as nodes.

Edge thickness, colour, label, dash, arrowhead — these are the weight and congruence channels. The reasoning content of the graph lives as much in the edges as in the nodes. Invest in edge visual quality with the same seriousness as node visual quality.

6 The viewer is a perception instrument, not a navigator.

Its job is to be transparent enough that the user forgets they are using it. Anything that draws attention to the viewer itself (lag, UI clutter, animation glitches, modal interruptions) competes with the perceptual act the user came to perform. Default toward calm, get-out-of-the-way design.

GUIDING QUESTIONS FOR FUTURE WORK

When proposing a change, ask:

- ☐ 1 Does this serve **simultaneous perception** of relationships, or does it force the user into a sequential read?

- ☐ 2 Does this preserve the **perceptual channel during motion**, or does it interrupt it?

- ☐ 3 Does this respect the **author's placement** as argument, or does it overwrite it?

- ☐ 4 Does this strengthen one of the three perceptual signals (**number, weight, congruence**), or does it add unrelated noise?

- ☐ 5 Does this make the viewer **more transparent**, or more demanding of the user's attention?

- ☐ 6 Does this **preserve the author's argument** when the file is seen by another viewer? Does it make the file easier or harder to share?

- ☐ 7 Does this stay useful **across sessions and over time**, or is it tied to a transient state?

If a change scores well on these, it probably belongs. If it scores poorly, it is probably the wrong design.